

INTELLIGENT INTERVIEW SIMULATION SYSTEM

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Problem Statment

Current interview preparation platforms provide limited realism and feedback,often relying on static questions and basic evaluation. They rarely assess verbal and non-verbal communication skills, creating a need for an AI-powered system that delivers realistic interview simulation and comprehensive candidate evaluation.

Proposed Solution

The proposed system is an AI-powered interview simulation platform that creates realistic interview experiences through an interactive virtual interviewer. It generates role-specific interview questions and evaluates candidate responses using AI-based speech, text, and behavioral analysis. The system provides structured feedback, enables natural conversational interaction, and supports recruiters with AI-driven candidate assessment and performance insights.

Project Objectives

- Provide a realistic AI-powered interview simulation environment.
- Help candidates improve their interview and communication skills.
- Analyze speech, tone of voice, facial expressions, and eye contact.
- Generate personalized feedback and performance reports.
- Ensure interview integrity through identity verification , eye tracking and multiface and phone detection.
- Offer a cross-platform solution for web/mobile for practcing use and desktop users for hiring use.

System Architecture

The Intelligent Interview Simulation System (IISS) uses a four-layer architecture consisting of the Frontend(Flutter), Backend(Node.js), Database(MongoDB), and AI Services layers(Python). It supports web, mobile, and desktop platforms, while AI modules handle interview generation, speech processing, behavioral analysis, and identity verification to deliver an intelligent and scalable interview experience.

Methodology

The proposed system conducts AI-driven interview sessions and performs real-time analysis of facial expressions, eye movements, speech content, and vocal characteristics. It also incorporates interview integrity mechanisms, including identity verification, eye tracking, multi-face detection, and mobile phone detection. The collected data are processed to evaluate candidate performance and generate personalized feedback. The system provides performance scores, evaluation reports, and improvement recommendations.

System Screens



Interview

25/05 | 03:53 | COMPLETED

36 / 100

NEEDS IMPROVEMENT

Performance breakdown

Communication

23%

Technical relevance

17%

Vocal confidence

18%

Non-verbal behaviour

22%

Summary

Below Average

The candidate struggled significantly, providing minimal or irrelevant answers. Communication clarity and technical depth were low, with frequent 'I don't know' responses. Vocal confidence was low, and non-verbal cues (looking away/up) suggested anxiety. However, no phone usage was detected, and face gate compliance was high.

Strengths

No phone or unauthorized device usage detected.

Maintained face presence throughout the interview.

Areas to improve

Very limited ability to answer technical questions, even at a basic level.

Low vocal energy and confidence, with many hesitations and incomplete answers.

Poor non-verbal behavior: predominantly looking away (up/right) instead of at the camera.

Lack of preparation for common full stack developer interview topics.

Action plan

Study core full stack concepts: MERN stack, data flow, authentication, and API design.

Practice answering questions out loud, recording yourself, and reviewing clarity.

Audio streaming

AI Agent

Question received

Let's go back.

End Interview

AI Agent

Question received

Let's go back.

Gamal

Main profile data

Subscription plan & transactions

Progress Monitoring

Overall Score

44

102

-21

Session

Date

Time

Report

Interview

14/06

10:16